OpenClaw API-KEY Configuration

1. Course Overview

This course will introduce in detail how to configure API-KEY for AI models in OpenClaw, including the API-KEY application process, two configuration methods (CLI command configuration and configuration file modification), and how to verify whether the configuration is successful. After completing this course, you will be able to successfully connect and use various AI large model service providers.

1.1 Course Content

- Understand the types of model service providers supported by OpenClaw
- Master the API-KEY application method
- Learn to view and verify the status of connected models

2. API-KEY Application

2.1 Recommended Model Service Providers

- **Recommended for beginners**: Refer to the accompanying tutorial [Chapter 11 Online AI Large Model Development] - [2. Tongyi Qianwen Large Model] to apply for the corresponding model provider API-KEY
- OpenClaw supports many model service providers, including:
  - **OpenAI** - GPT series (API + Codex)
  - **Anthropic** - Claude series (API + Claude Code)
  - **Google** - Gemini series

    - **Alibaba Cloud** - Qwen Cloud, Tongyi Qianwen
    - **DeepSeek** - DeepSeek model
    - **Zhipu AI** - GLM model
    - **Moonshot** - Kimi + Kimi Coding
    - **Ollama** - Local model (cloud + local)
    - And many other service providers
- You can freely choose a model service provider based on your actual needs and regional network policies, and register and apply for API-KEY on the corresponding official website

3. Configure OpenClaw API-KEY

3.1 Configuration Methods

OpenClaw provides two methods for configuring API-KEY:

1. **CLI Command Configuration** (suitable for beginners)
2. **Direct Configuration File Modification** (suitable for developers)
    - Configuration file path: `$HOME/.openclaw/openclaw.json`

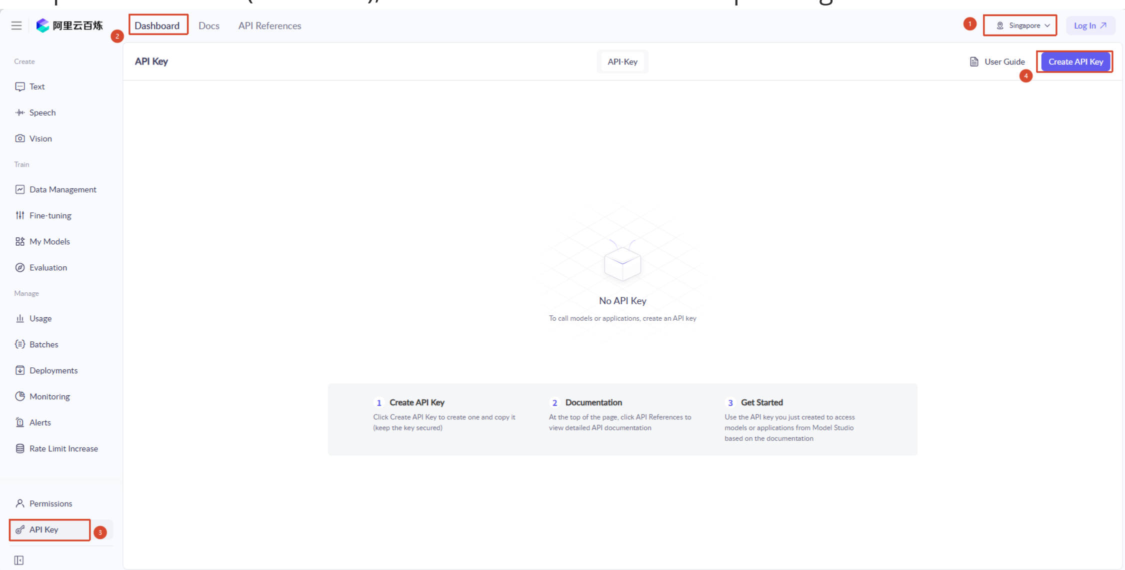
3.2 Quick Configuration

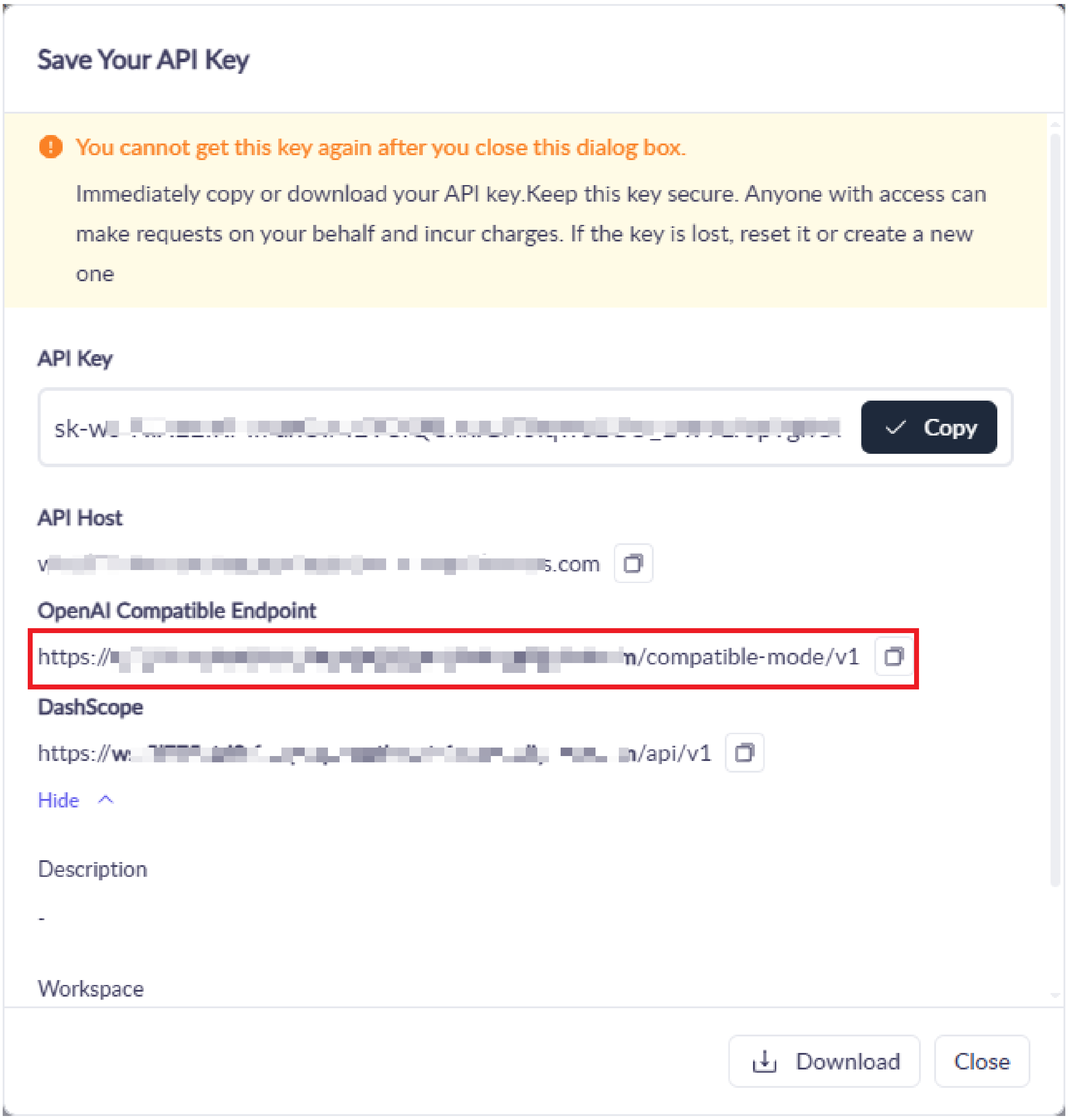
- Suitable for users who want to quickly get started. You need to first register for an Alibaba Cloud Bailian large model account and create an API-KEY (refer to the accompanying tutorial [Chapter 11 Online AI Large Model Development] - [2. Tongyi Qianwen Large Model])
- The factory default configuration file already has Bailian model parameters configured. Simply fill in your API-KEY and baseUrl to start using it directly.
- Open OpenClaw configuration file:

```
nano $HOME/.openclaw/openclaw.json
```

3.2.1 Configure baseUrl

- Users need to configure the `tongyi_base_url` corresponding to their account. Go to the [Model Service Platform Bailian Console](#) to apply for an API key. Copy the API key and OpenAI-compatible address (Base URL), then fill them into the corresponding fields.





3.2.2 Fill in apiKey and baseUrl

```
"models": {
  "mode": "merge",
  "providers": {
    "bailian": {
      "baseUrl": "https://dashscope.aliyuncs.com/compatible-mode/v1",
      "apiKey": "sk-a87e1137b17b4a35af13aa4a12ac3473",
```

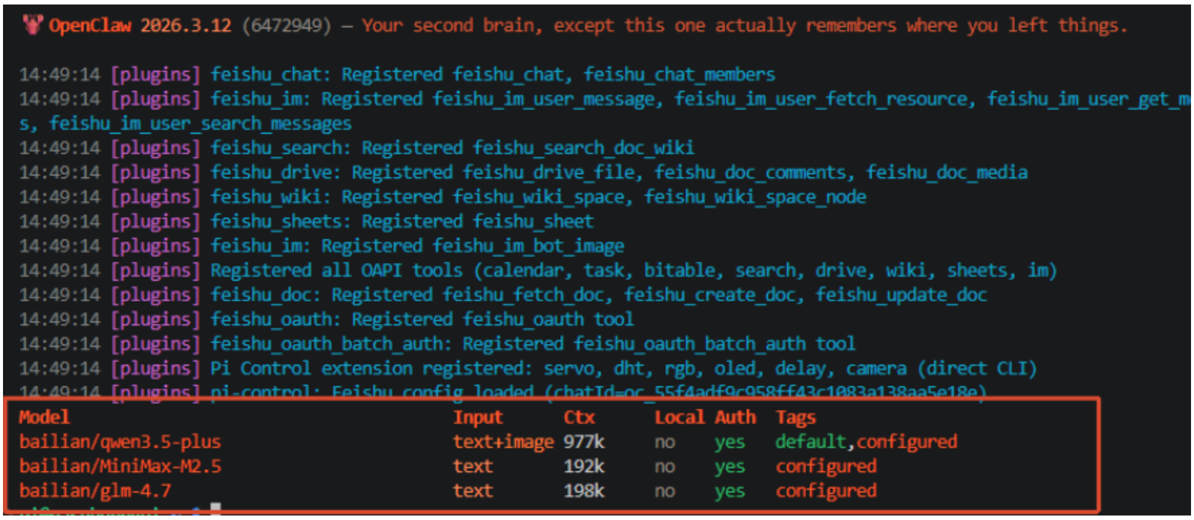
- After filling in, press `Ctrl+X`, then press `Y` to confirm save, then press Enter to exit

4. View Connected Model Status

4.1 View Configured Model List

After configuration is complete, you can use the following command to view connected models:

```
openclaw models list
```

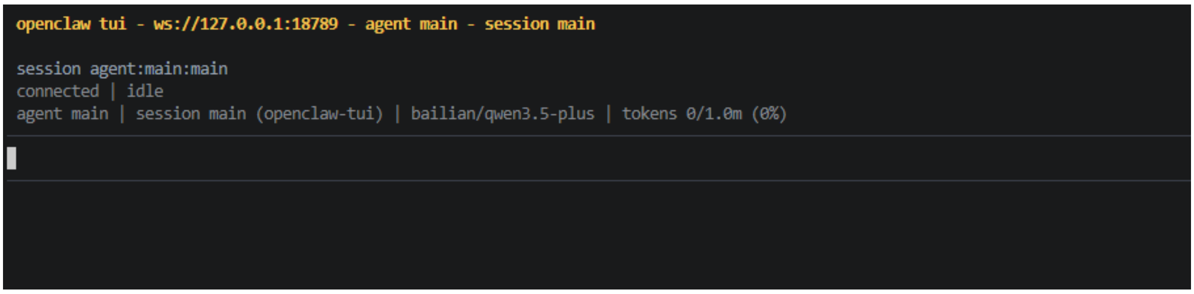


4.2 Verify Whether Configuration is Successful

1. **Test Dialogue**

```
openclaw tui
```

Start the TUI interface, send a test message (such as "Hello"), confirm the model can respond normally.



5. Other Access Methods (Advanced)

- If you have your own model provider or need to connect to other model providers later, refer to the following tutorial

5.1 CLI Command Configuration API-KEY (Suitable for Beginners)

1. **Start Configuration Wizard**

```
openclaw onboard
```

- Select【Yes】, press Enter

```
Recommended baseline:
- Pairing/allowlists + mention gating.
- Multi-user/shared inbox: split trust boundaries (separate
  gateway/credentials, ideally separate OS users/hosts).
- Sandbox + least-privilege tools.
- Shared inboxes: isolate DM sessions (`session.dmScope:
  per-channel-peer`) and keep tool access minimal.
- Keep secrets out of the agent's reachable filesystem.
- Use the strongest available model for any bot with tools or
  untrusted inboxes.

Run regularly:
openclaw security audit --deep
openclaw security audit --fix

Must read: https://docs.openclaw.ai/gateway/security

◆ I understand this is personal-by-default and shared/multi-user use requires
lock-down. Continue?
  ● Yes / ○ No
```

- Select QuickStart for configuration mode, press Enter

```
◆ Setup mode
  ● QuickStart (Configure details later via openclaw configure.)
  ○ Manual
```

2. **Select Model Service Provider**
    - Use up/down arrows to select the service provider for which you have applied for API-KEY, press Enter to confirm. Here we take Alibaba Cloud Bailian model provider as an example. The configuration method for other model providers is the same
    - Here select custom provider **【Custom Provider】**

```
◆ Model/auth provider

  Search:
  ○ Alibaba Model Studio
  ○ Anthropic
  ○ Arcee AI
  ○ BytePlus
  ○ Chutes
  ○ Cloudflare AI Gateway
  ○ Copilot
  ● Custom Provider (Any OpenAI or Anthropic compatible endpoint)
```

    - Fill in the model provider's base_url

```
◆ API Base URL
  https://dashscope.aliyuncs.com/compatible-mode/v1
```

3. Input API-KEY

- After selecting the service provider, the system will prompt you to enter the corresponding API-KEY

```
◆ How do you want to provide this API key?
│ ● Paste API key now (Stores the key directly in OpenClaw config)
│ ○ Use external secret provider
└
```

- Paste or enter the API-KEY you applied for, press Enter to confirm

```
◆ API Key (leave blank if not required)
│ sk-a87e1137b17b4a35af13aa4a1xxxxxxx
└
```

- Select interface protocol **OpenAI-compatible**

```
◆ Endpoint compatibility
│ ● OpenAI-compatible (Uses /chat/completions)
│ ○ Anthropic-compatible
│ ○ Unknown (detect automatically)
└
```

4. Fill in model name, here we take MiniMax as an example

```
◆ Model ID
│ MiniMax-M2.5
└
```

- **Endpoint ID** is the model provider identifier, can be filled in freely

```
◆ Endpoint ID
│ custom-dashscope-aliyuncs-com
└
```

- Model alias (optional) leave blank directly

```
◆ Model alias (optional)
│ e.g. local, ollama
└
```

5. **Other Configurations**

- For other configuration items, select skip as shown below, no configuration needed

```
◇ Skills status

  Eligible: 16
  Missing requirements: 38
  Unsupported on this OS: 7
  Blocked by allowlist: 0


◆ Configure skills now? (recommended)
  ○ Yes / ● No
```

- For Enable Hooks, press space to select **Skip for now**

```
Enable hooks?
■ Skip for now
□ 🚀 boot-md
□ 📎 bootstrap-extra-files
□ 📝 command-logger
□ 💾 session-memory
```

- Select Restart to restart gateway and apply configuration

```
Gateway service already installed
● Restart
○ Reinstall
○ Skip
```

5.2 Modify Configuration File (Recommended Method)

Configuration File Location

Configuration file is located at: `$HOME/.openclaw/openclaw.json`

Edit Configuration File

1. **Open Configuration File**

```
nano $HOME/.openclaw/openclaw.json
# Or use your preferred editor, such as vim, code, etc.
```

2. **Add Model Provider Configuration**

Add or modify the `models.providers` section in the configuration file:

```
{
  "models": {
    "providers": {
      "openai": {
        "apiKey": "OPENAI_API_KEY"
      },
      "anthropic": {
        "apiKey": "ANTHROPIC_API_KEY"
      },
      "google": {
        "apiKey": "GOOGLE_API_KEY"
      }
    }
  },
  "agents": {
    "defaults": {
      "model": {
        "primary": "openai/gpt-4"
      }
    }
  }
}
```

3. **Configuration Description**
    - `models.providers`: Configure API-KEY for each model service provider
        - `openai.apiKey`: OpenAI's API-KEY
        - `anthropic.apiKey`: Anthropic's API-KEY
        - `google.apiKey`: Google's API-KEY
    - `agents.defaults.model.primary`: Set the default model to use
        - Format is `provider/model-name`, such as `openai/gpt-4`, `anthropic/claude-3`, `google/gemini-pro`
4. **Save and Exit**
    - Press `Ctrl+O` to save, `Ctrl+X` to exit (nano editor)
5. **Factory Default Bailian Configuration (can be directly modified or referenced)**

```
"models": {
    "mode": "merge",
    "providers": {
    "bailian": {
      "baseUrl": "https://dashscope.aliyuncs.com/compatible-mode/v1",
      "apiKey": "sk-a98b519b6bab4e35a7b82b4a95b8a4f6",
      "api": "openai-completions",
      "models": [
        {
          "id": "qwen3.5-plus",
          "name": "qwen3.5-plus",
          "reasoning": false,
          "input": [
            "text",
            "image"
```

```
        ],
        "contextWindow": 1000000,
        "maxTokens": 65536,
        "cost": {
          "input": 0,
          "output": 0,
          "cacheRead": 0,
          "cacheWrite": 0
        }
      },
      {
        "id": "MiniMax-M2.5",
        "name": "MiniMax-M2.5",
        "reasoning": false,
        "input": [
          "text"
        ],
        "cost": {
          "input": 0,
          "output": 0,
          "cacheRead": 0,
          "cacheWrite": 0
        },
        "contextWindow": 196608,
        "maxTokens": 32768
      },
      {
        "id": "glm-4.7",
        "name": "glm-4.7",
        "reasoning": false,
        "input": [
          "text"
        ],
        "cost": {
          "input": 0,
          "output": 0,
          "cacheRead": 0,
          "cacheWrite": 0
        },
        "contextWindow": 202752,
        "maxTokens": 16384,
        "compat": {
          "thinkingFormat": "qwen"
        }
        }
    ]
  }
```

6. Common Problems and Solutions

6.1 API-KEY Invalid or Error

Problem: Configuration prompts API-KEY is invalid or authentication failed

Solution:

1. Confirm whether the API-KEY is correct (note no extra spaces)
2. Check if the API-KEY has expired or been disabled
3. Confirm whether the account balance is sufficient (some service providers require pre-deposit)

6.2 Unable to Access Model Service

Problem: Configuration is correct but cannot connect to model service

Solution:

1. Check if network connection is normal
2. Confirm whether your regional network can access the service provider (some services may require special network configuration)
3. Try changing to other model service providers for testing

7. Course Summary

Through this course, you have mastered:

- [x] Types of model service providers supported by OpenClaw
- [x] API-KEY application method and precautions
- [x] CLI command configuration method (suitable for beginners)
- [x] Configuration file modification method (suitable for developers)
- [x] Methods to view and verify model configuration status
- [x] Troubleshooting and solutions for common problems